VIT®
CHENNAI
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

MITRE ATT&CK Threat Intelligence Mapping System

SUBJECT : FUNDAMENTALS OF CYBER SECURITY

SUBJECT CODE : BCSE338L

SLOT : G2 + TG2

FACULTY : Dr. Prasad M

SUBMITTED BY :

R Harish — 24BYB1159

Harshitha Sajith Menon — 24BYB1166

Priyadharshini S — 24BYB1068

1. Introduction

The rapid expansion of digital technologies and internet connectivity, cyber attacks have especially increased in terms of both complexity and frequency. Global organizations are continuously under threat from attackers who use a wide spectrum of techniques, malware, and tools to gain access to systems and steal confidential data. Therefore, a thorough understanding of how attackers operate and analyzing their attack methods is a necessity in nowadays cybersecurity.

MITRE ATT&CK framework is a knowledge base extensively used that essentially captures the behaviors of real-world adversaries; namely, it records attack tactics, techniques, and procedures (TTPs). It makes available a detailed and organized description of how attackers break into a system, perform harmful acts, stay undetected, and communicate with compromised systems. Security professionals frequently use this framework as a tool to dissect the nature of attacks and to devise and reinforce defenses.

Nevertheless, the bulk of data contained in the MITRE ATT&CK framework may pose a challenge to users who want to easily discover and analyze the links between threat actors, malware tools, and attack techniques. To delve into this data efficiently, a system capable of arranging and showing cyber threat intelligence data in a well-ordered and user-friendly way is indispensable. This paper describes a Cyber Threat Intelligence (CTI) Platform that utilizes the MITRE ATT&CK framework and assists users in exploring and analyzing threat intelligence data thanks to an interactive dashboard.

On this platform, the user can study various threat entities like threat, actors, malware tools, and techniques while at the same time mapping the different stages of an attack and the relations among these entities. By bringing together all pertinent information in one interface, the platform certainly enables users to gain a deeper insight into how attackers behave and the anatomy of cyber attacks.

2. Problem Statement

The nature of modern cyber attacks is that they are becoming more and more sophisticated and they are carried out by threat actors who are capable of utilizing a large array of malware, hacking tools, and visitations techniques to of systems and networks. Security analysts heavily depend on threat intelligence platforms such as the MITRE ATT&CK framework that offers thorough details about the tactics, techniques, and procedures of the enemy that have been witnessed in real attacks. To be more specific, the MITRE ATT&CK framework is a very powerful tool for figuring out the behavior of an attacker. However, the large number of structured threat intelligence data makes it difficult for the users to effectively study and understand the interrelations between different threat entities.

Most of the time, security analysts have to manually look through several datasets and documents to find out how threat actors are linked to specific malware, tools, and attack techniques. Besides, even the existing platforms, which are supposed to present threat intelligence data often in fragmentation, make it difficult to get an overall view of the whole attack ecosystem, as they are still presenting the information in a fragmented manner. What is more, this can be a time-consuming process and even that might slow down threat investigation and analysis.

For that reason, the usage of one integrated solution that is capable of making the work easier in conducting threat intelligence investigations by bringing related information in the form of a structured and interactive manner will be very much needed. This system should be able to give users the possibility of discovering the ways by which threat actors, malware tools, and techniques are related to each other, making it easy for them to visualize the attack patterns and to conduct the analysis of the behavior of adversaries all in one platform.

This platform allows users to check different cyber threats through an engaging dashboard that unites threat intelligence data and enables the users to visually see the techniques of the attackers and the phases of the attack. By arranging all the intelligence that is related to a threat on a single page.

3. Proposed Solution

The present system is intended to be a Cyber Threat Intelligence Platform that can greatly assist in the discovery and analysis of cyber threat data in accordance with the MITRE ATT&CK framework. It features a user-friendly online dashboard through which users can examine the connections between various types of threat entities, e.g. threat actors, malware tools, and attack techniques.

It starts with processing the threat intelligence data from the MITRE ATT&CK dataset and then formatting the data in a coherent manner. When a user enters the name of a particular threat entity, the platform will not only fetch the required data but also present all related intelligence visually in one screen. This way, users are able to grasp very fast the interrelations between the components of a cyber attack.

Our system is a Flask-based backend that receives and processes user requests for threat intelligence. This backend communicates with the AttackMatrix engine, which is a MITRE ATT&CK parser that pulls out details of threat actors malware tools techniques etc. The resulting information is delivered to the frontend dashboard for display in a user-friendly and visually appealing way.

The web interface provides a platform to navigate threat intelligence through a single dashboard that highlights critical data such as threat explanation, corresponding techniques, attack stages, and the connections among various threat entities.

Besides just visual interpretation, the platform creates structured threat intelligence reports that encapsulate concise information related to specific threats. Such reports assist users in recording their investigative work and provide them with insights into the various attack strategies deployed by adversaries.

The suggested solution combines the structured threat intelligence data with a user-friendly dashboard and visualization features, thereby assisting cybersecurity researchers, analysts, and students in effectively exploring cyber threats as well as studying the behavior of attackers through the lens of the MITRE ATT&CK framework.

4. Objectives

The project is focused on creating a Cyber Threat Intelligence Platform that enables the easy discovery and analysis of cyber threats according to the MITRE ATT&CK framework. The platform is designed to offer a well-structured and user-friendly space where one can get information about threat actors, malware tools, and attack techniques as well as investigate their interconnections.

- Designing a platform that allows users to access cyber threat intelligence online.
- Fetching the MITRE ATT&CK data and converting information on threat actors, malicious software, tools, and techniques for platform use.
- Offering a consolidated workspace whereby different threat entities are accessible to users and they can observe the relationships between entities.
- Representing attack techniques as well as attack phases with graphical charts and dashboards.
- Demonstrating the attack flow and the behavior of adversaries by means of an attack timeline.
- Presenting to users all relevant intelligence information in a single, integrated interface in order to facilitate their fast analysis of cyber threats.
- Creating well-structured threat intelligence reports that outline the research results.
- Providing a tool to cybersecurity students, researchers, and analysts to observe enemy behavior and attack methods through the MITRE ATT&CK framework.

5. System Architecture

This system is based on a modular client-server architectural model. It is designed to take threat intelligence data from the MITRE ATT&CK framework and display it via a user-friendly web interface. The architecture is organized around three primary components: the User Interface Layer, the Application Processing Layer, and the Threat Intelligence Data Layer.

5. 1. User Interface Layer

The User Interface Layer is the part of the software that you as a user communicate with. It shows threat intelligence data and uses the web technologies HTML, CSS, and JavaScript. The interface is like a web dashboard through which users can locate threat entities such as malware, threat actors, tools, and techniques.

This layer displays:

- Threat intelligence details
- Attack technique tables
- Attack phase distribution charts
- Threat severity indicators
- Attack timeline visualization

JavaScript is used to handle user interactions, dynamic content updates, and chart visualizations using Chart.js.

5. 2. Application Processing Layer

The Application Processing Layer is based on the Flask web framework and is written in Python. This layer serves as the heart of the system and is responsible for executing all the processing tasks involved in threat intelligence analysis.

The main responsibilities of this layer include:

- Handling user search requests
- Processing threat intelligence queries

- Extracting relevant information from the MITRE ATT&CK dataset
- Organizing relationships between threat actors, malware, tools, and techniques
- Generating threat intelligence reports
- Preparing data for visualization on the dashboard

The Flask application processes the data and sends the structured results to the frontend for display.

5. 3. Threat Intelligence Data Layer

The Threat Intelligence Data Layer stores the cyber threat intelligence that is structurally represented and extracted from the MITRE ATT&CK dataset. The dataset undergoes processing with the AttackMatrix engine, which is responsible for downloading and arranging the MITRE ATT&CK data in an orderly manner.

This layer stores information related to:

- Threat actors
- Malware
- Attack tools
- Attack techniques
- Attack matrices
- Relationships between threat entities

The AttackMatrix engine processes the dataset and provides the required information to the application layer when a user performs a search or investigation.

5. 4. Visualization and Reporting Component

The platform comes with a visual representation element that graphs the threat intelligence data. The use of Chart.js enables the creation of charts and other types of visual clues that show the distribution of the attack phases and provide other forms of analytical insights.

Besides that, the platform produces PDF documents that highlight threat intelligence results. These documents contain organized data about threats, the techniques used, and insights about defense.

User Search Query

↓

Web Interface (HTML/CSS/JavaScript)

↓

Flask Backend (Application Processing Layer)

↓

Threat Intelligence Processing

↓

MITRE ATT&CK Dataset (AttackMatrix)

↓

Processed Intelligence Data

↓

Visualization & Dashboard Output

↓

Optional PDF Report Generation

6. Technology Stack

The Cyber Threat Intelligence Platform is built with a variety of programming languages, frameworks, and tools, which allow for the smooth processing, visualization, and making threat intelligence data understandable. The system uses both sides technologies: the server side (backend) and client side (frontend) to make an interactive and easy to use dashboard for the analysis of cyber threats by using the MITRE ATT&CK framework.

6. 1.Programming Language

Python

Python stands as the backbone language for implementing the system's backend. It carries out the tasks of processing the threat intelligence data, managing user interactions, and accessing the MITRE ATT&CK dataset. One of the reasons why Python was selected is because of its excellent features supporting data processing, web development, and cybersecurity applications.

6. 2. Web Framework

Flask

The team opted for Flask as the Python web framework to construct the backend web application because of its simplicity. Flask will take care of HTTP requests routing user input processing, and finally linking the backend workings with the frontend dashboard.

6. 3. Frontend Technologies

HTML

HTML is the fundamental building block of the web interface and is used to develop the dashboard's visual formatting.

CSS

CSS is the technology used to beautify the application, e. g. by positioning, coloring, and generally styling the dashboard elements. JavaScriptJavaScript powers the frontend's interactivity by enabling features such as interactive dashboards, pop-up windows, and user interaction processing.

6. 4. Visualization Library

Chart.js

Chart.js is a charting library written in JavaScript that is widely used for producing various chart types including distribution charts for attack phases. Such charts aid in communicating threat intelligence data efficiently to the users.

6. 5. Threat Intelligence Framework MITRE ATT&CK Framework

The MITRE ATT&CK framework is considered as the main repository of cyber threat intelligence data for the system. It captures in detail the tactics, techniques, and procedures (TTPs) of attackers as well as tools and behaviors of adversaries that have been recorded in actual cyber incidents.

6. 6.Threat Intelligence Processing Engine AttackMatrix

AttackMatrix is a tool that is used for parsing the MITRE ATT&CK dataset to identify and extract the most pertinent information. It assists in structuring threat intelligence data and making it possible for the system to locate connections between threat actors malware tools, and techniques.

7. Key Features

7. 1. MITRE ATT&CK Threat Intelligence Integration

The system connects seamlessly with the AttackMatrix service which supplies formatted threat intelligence data directly via the MITRE ATT&CK dataset.

When a user searches for a threat (malware, actor, etc.), the system retrieves detailed information such as:

- techniques used
- associated threat actors
- attack tools
- MITRE references

7. 2. Malware Search Engine

The program has a feature that enables users to type the name of a malware or a threat entity for finding relevant information quickly.

The backend:

- sends a request to the AttackMatrix API
- retrieves the threat intelligence data
- extracts techniques, actors, tools, and descriptions
- displays them on the dashboard.

This allows users to quickly investigate cyber threats.

7. 3. Malware Catalog Browser

The software consists of a malware catalog that refreshes itself from time to time by fetching new data from the MITRE dataset without any requirement of user intervention.

With it, users can check out the malware directory choose malware from the set automatically fill in the search box

Consequently, the search for various threats is made quite simple and the user is not expected to be familiar with the names of malwares already.

7. 4. Threat Intelligence Dashboard

Threat intelligence data is shown on a highly organized panel.

The dashboard presents:

- threat name
- STIX identifier
- number of techniques used
- malware description
- associated actors
- attack tools
- techniques table

This provides a centralized view of threat intelligence data.

7. 5. Attack Phase Visualization

The application generates a doughnut chart using Chart.js that shows the distribution of attack techniques across different phases.

Example phases:

- execution
- credential access
- initial access
- defense evasion
- other

This visualization helps analysts quickly understand attacker behavior patterns.

7. 6. Attack Kill Chain Timeline

Based on the detected techniques, the timeline auto-generates a path through the attack phases. The sequence reveals when actions happen, one step after another, and **how far along the attacker is.**

7. 7. Interactive Intelligence Popups

You click any item like technique, actor, tool, or malware to see what's behind it, descriptions appear instantly in a popup window. No clutter. Just clean data you need when you want it.

7. 8. Threat Severity Assessment

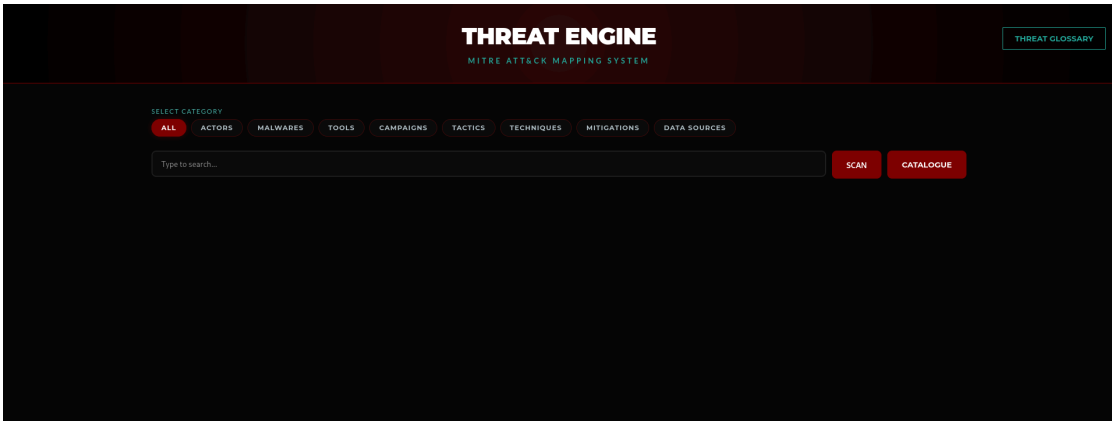
The risk score grows with every new technique added, the more methods used, the higher the threat level becomes.

7. 9. Automated Threat Intelligence Report Generation

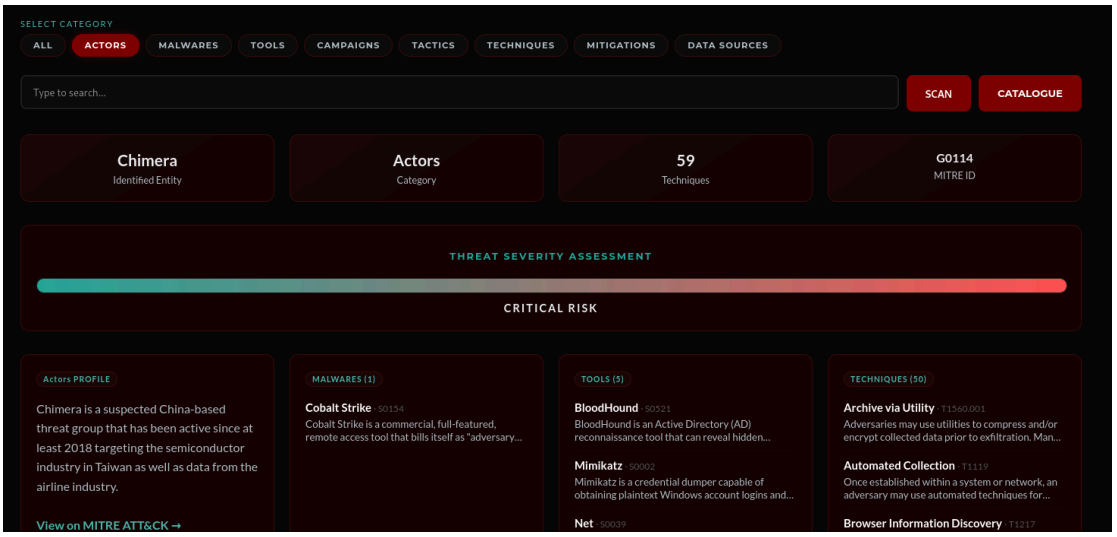
The system generates a PDF report with malware names, attack techniques, detailed descriptions, and defense data - no fluff, just solid facts. The user saves findings for records or deeper study without lifting a finger.

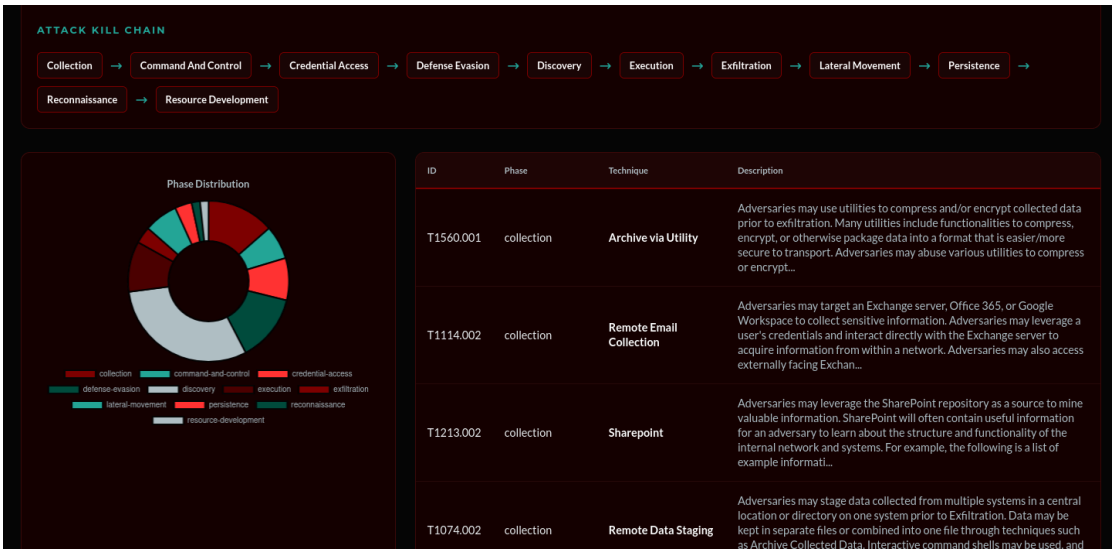
Reports come packed with specifics: what attacks happened, how they work, and exactly what defenses were activated.

8. Output

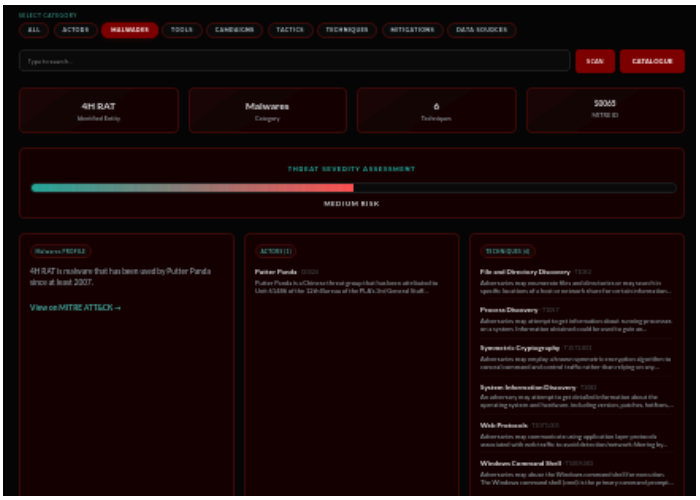


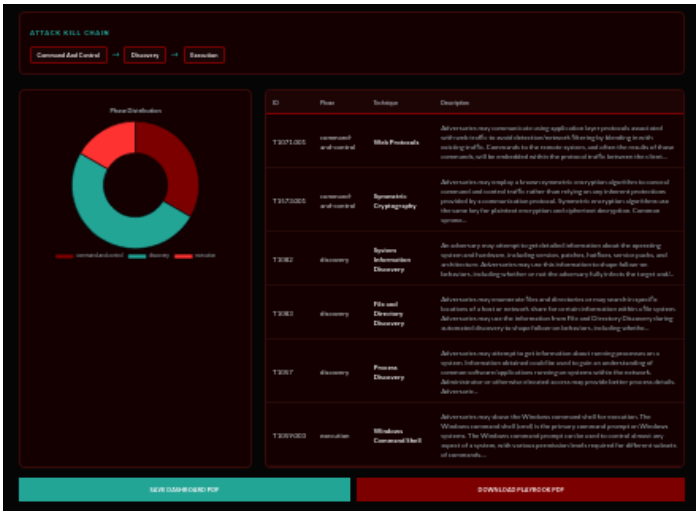
8. 1. Actors



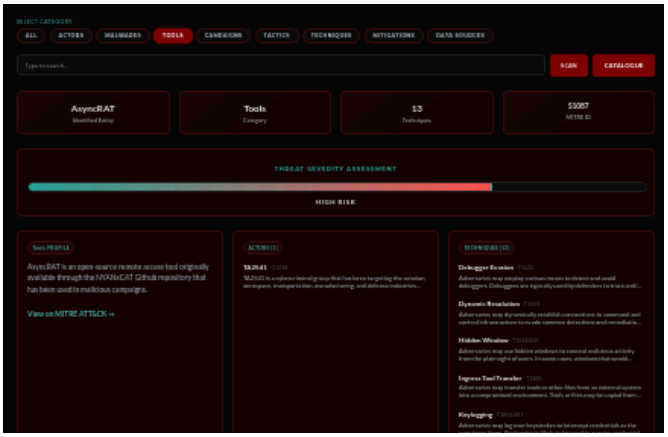


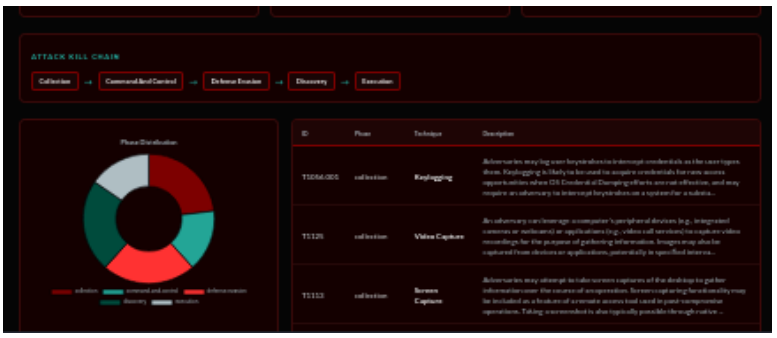
8. 2. Malwares



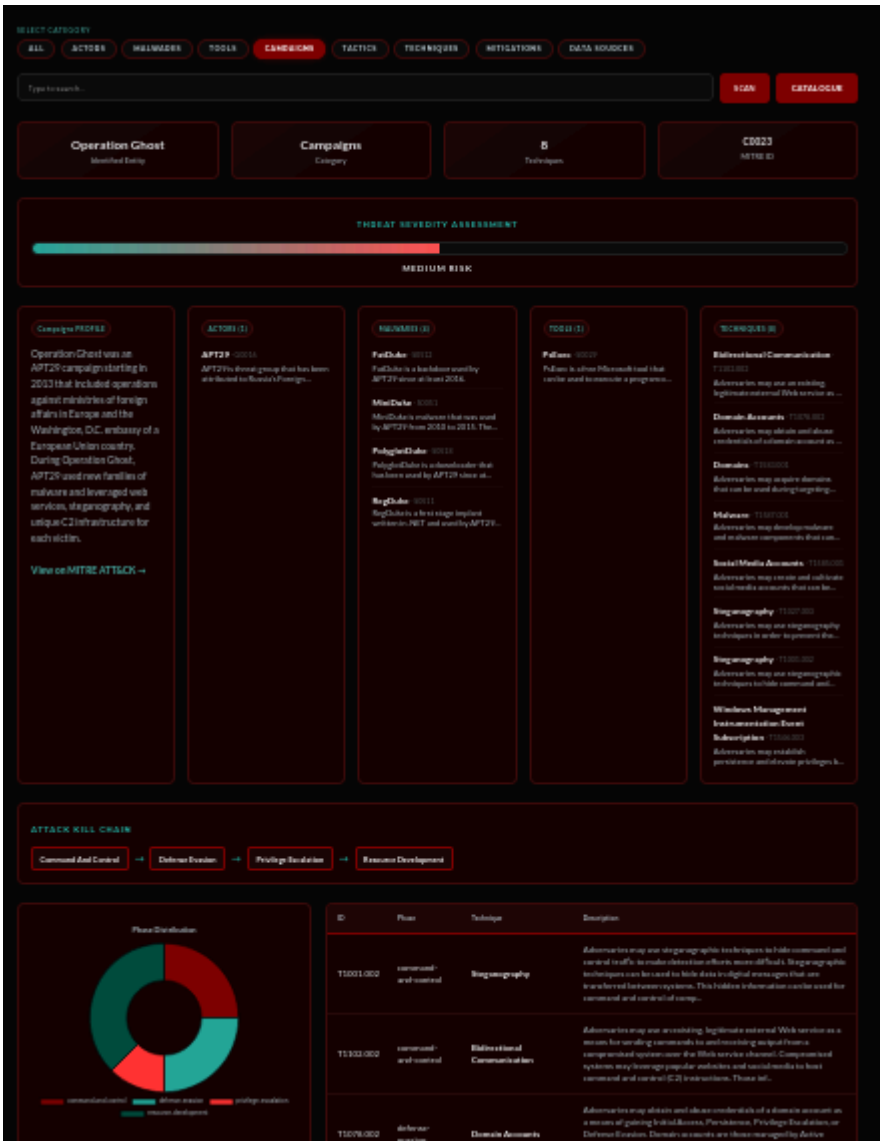


8. 3. Tools

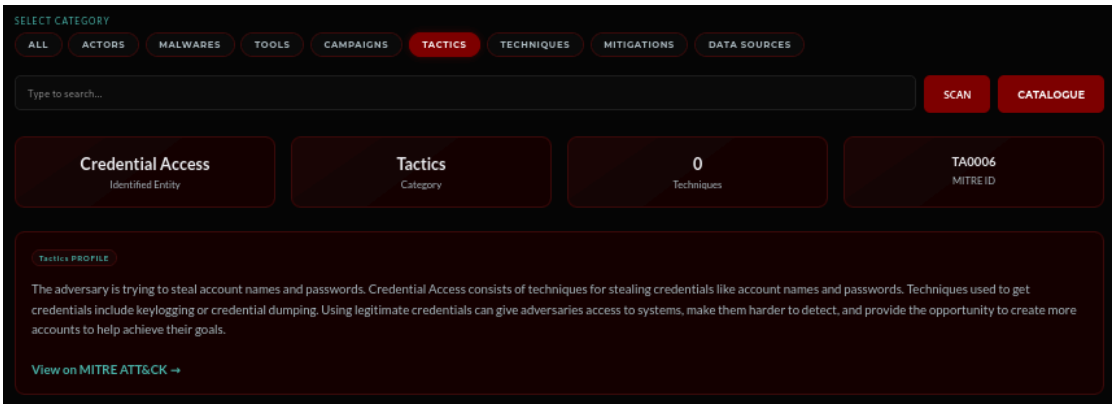




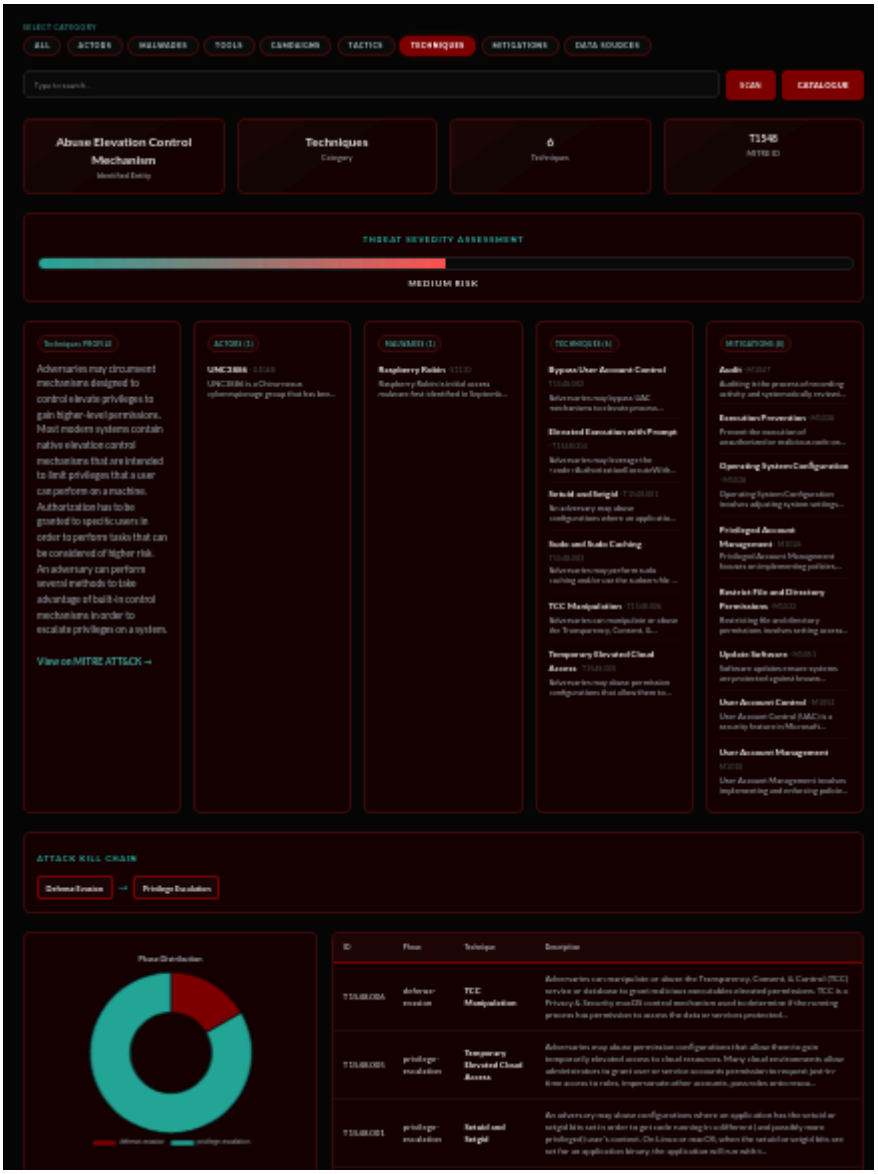
8. 4. Campaigns



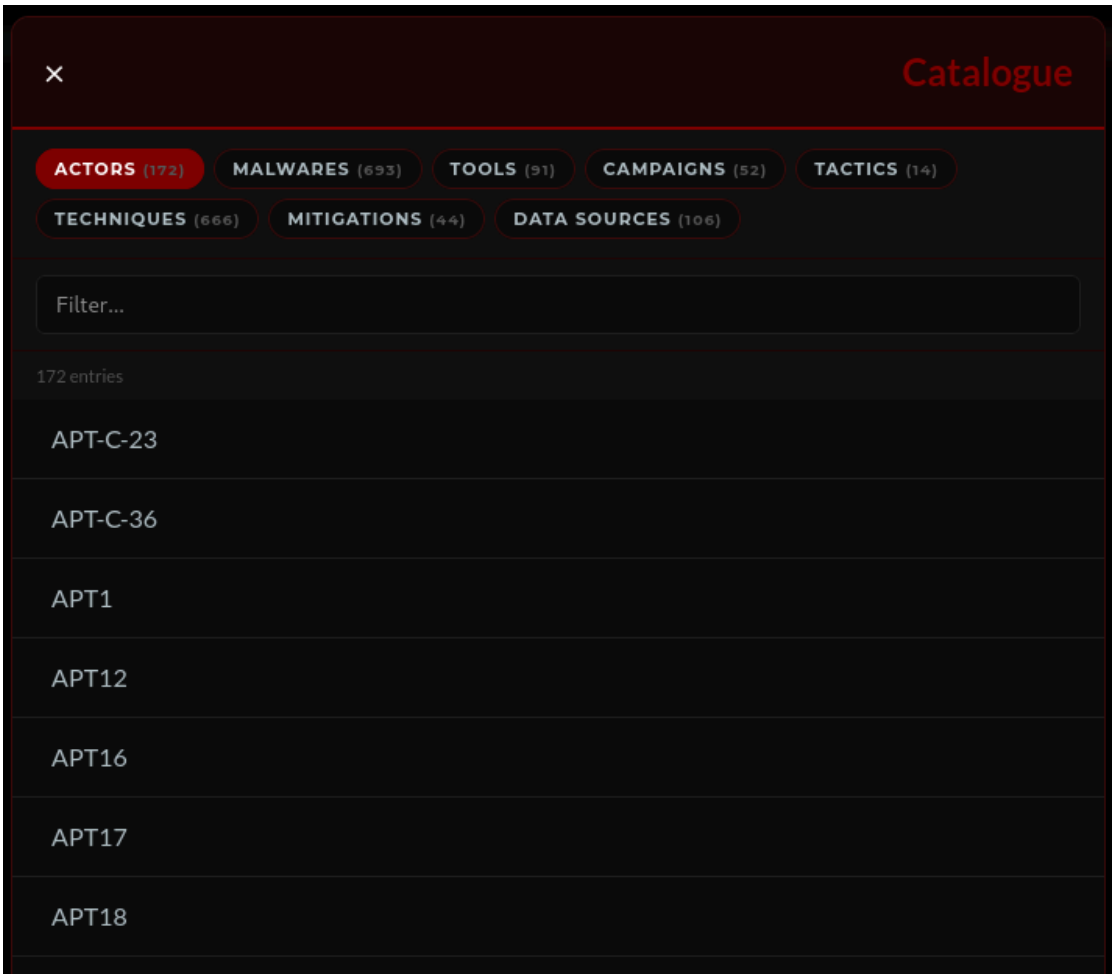
8. 5. Tactics



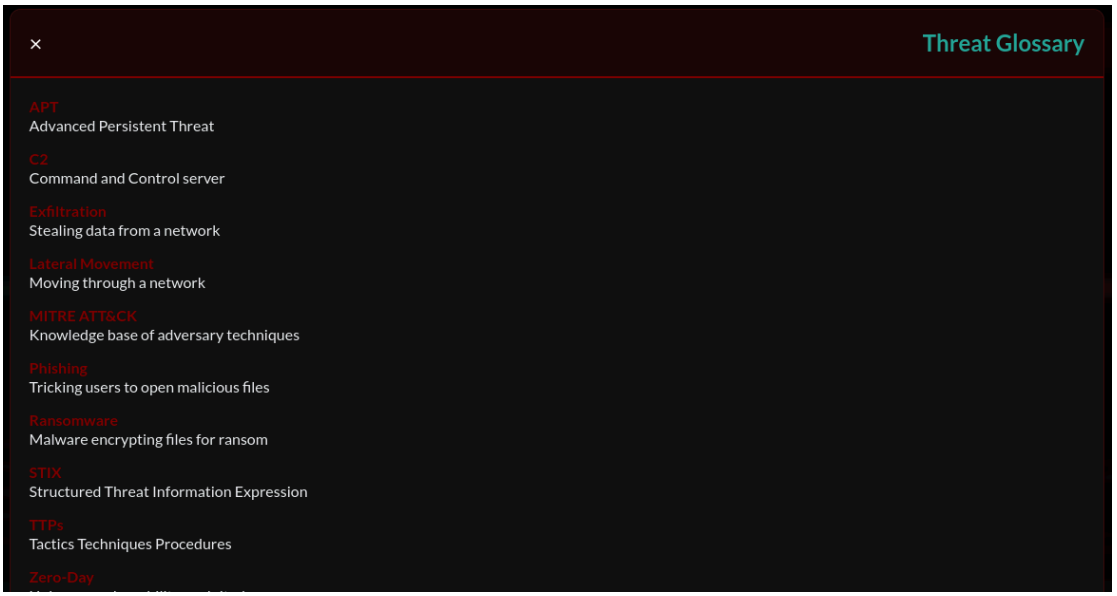
8. 6. Techniques



8. 7. Catalogue



8. 8. Threat glossary



8. 9. Options

Additional features such as options to save the dashboard and download the playbook are present at the end of each playbook.

9. Applications

Cyber Threat Intelligence Platform is a new tool proposed that can be implemented across many different cybersecurity-related areas, serving as a means to thoroughly analyze and comprehend cyber threats through the latest cyber attack data collection and analysis methods.

9. 1. Security Operations Centers (SOC)

Security Operations Center teams can leverage this system for rapid threat analysis by accessing the information on malware, techniques, and attack patterns offered by the MITRE ATT&CK framework.

9. 2. Cybersecurity Research

With this platform, researchers will be able to examine adversary behavior, attack techniques, and relationship between threats in a hierarchical and easy-to-comprehend way.

9. 3. Educational Purpose

Besides providing comprehensive knowledge about cybersecurity, the system also serves as a practical guide through systematized cyber attacks.

9. 4. Incident Analysis

Security analysts can comprehend with the help of the system additional incident details such as the origin of an attack, the weaponry used, and the sequence of attack events.

9. 5. Threat Intelligence Analysis

The toolset underneath the platform can provide useful leverage to analysts in following and discovering relations among the elements:

malware components threat actors, and techniques.

Through this platform, a user may have better and greater overall insight into threat intelligence.

9. 6. Cybersecurity Training & Awareness

Enterprises and bureaucracies may utilize the platform as a means to provide personnel and teams with knowledge of attacker behavior along with the scenarios of recent attacks.

10. Conclusion

The Cyber Threat Intelligence Platform that we propose illustrates well that NOT only can structured threat intelligence data from the MITRE ATT&CK framework be turned into information that is ready for presentation and visualization through an interactive web dashboard but also how efficiently it can be done.

The users of the platform are given the means to uncover malicious cyber activity in an intuitive interface by providing them not only with information on the malware but also on techniques used, actors, and tools involved in an attack.

The combination of the AttackMatrix engine with a Flask-based backend enables the system to be capable of quickly fetching and arranging threat intelligence data. Together, the interactive dashboard and visual components (e.g. charts and an attack timeline) allow the user to gain a clear understanding of the attacker's behavior and the phases of a cyber attack. Having clickable elements and more detailed pop-up windows gives users the option to more deeply engage with the threat intelligence that is presented without introducing any complexity. Additionally, the system can

generate reports thus it can work as a handy writing and analytical tool in the hands of users. To summarize, the platform facilitates the normally daunting task of revelation and exploration of intricate cybersecurity data and stands as a useful tool that may be used by students, researchers, and security analysts for studying and understanding the attack methodologies commonly used in the real world.

In summary, the project delivers what it set out to do, that is, build a highly efficient and very easy to use threat intelligence analysis system. Besides that, it also shows how critical visualization and structured data are in raising cyber security awareness.